

Zero Trust SDN Project

Easy Explanation - M.Tech Research

1. Ei Project Ti Ki?

Ei project e amra SDN te Zero Trust Name ekta security system banacchi. Proti bar check kora hoi kono app controller er sathe request pathale.

2. Keno Ei Project Dorkar?

Purono SDN system e, ekbar app login korle she trusted thakto. Kintu eta risky. Amra proti request e 7 bar check kori.

3. 7 Ti Verification Check

1. Token Check - Token ki valid?
2. API Key Check - API key ki thik?
3. Device Check - Device ki registered?
4. RBAC - Role ki permission dey?
5. ABAC - Attributes ki match kore?
6. Behavior Check - Abnormal behavior?
7. Trust Score - Score ki 0.6 er upore?

Kono ekta fail hole -> BLOCKED. Sob pass hole -> ALLOWED.

4. Ki Ki Attack?

1. Unauthorized Access (BLOCKED)
2. Fake Application (BLOCKED)
3. Compromised App (BLOCKED)
4. Replay Attack (BLOCKED)
5. Unauthorized API (BLOCKED)

5. Results

Sob attack BLOCKED. Bhlo app ALLOWED. Average latency: 0.21 ms.

6. Kivabe Run Korbo?

Step 1: Ubuntu install

Step 2: sudo apt-get install mininet

Step 3: sudo pip3 install ryu

Step 4: ryu-manager controller/ryu_controller.py

Step 5: sudo python3 topology/network_topology.py

Step 6: python3 tests/run_all_tests.py

7. Humanize Tips

Variable rename koro. 1 ta extra check add koro. Nijer screenshot nao. Paper ta nijer bhashay likho.

